

Rose Carcaillet

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PROFILE

Aerospace engineering graduate (BEng, QMUL), incoming MSc Aeronautical Engineering student at Imperial College London. Background in orbital mechanics and space systems, with a growing interest in AI safety and governance as applied to space infrastructure.

EDUCATION

MSc Advanced Aeronautical Engineering, Imperial College London	<i>2026 – 2027</i>
BEng (Hons) Aerospace Engineering, Queen Mary University of London Achieved 1st – 78.2%	<i>2023 – 2026</i>
French Baccalauréat – Highest Honours, 18.48/20	<i>2022</i>
A-Levels (One-Year Course) – AAA (Physics, Biology, Chemistry)	<i>2023</i>

RESEARCH PROJECT

BEng Dissertation — Improving Space Debris Orbit Prediction Using Space Debris Laser Ranging (SDLR) *April 2026*

- Inspired by the first ESA-sponsored Space Debris Hackathon at Technische Universität Berlin; retained dataset access in partnership with DiGOS Potsdam GmbH to investigate orbit determination (OD) for uncooperative LEO debris via single-station SDLR, applying batch least-squares estimation in Python (Orekit) with full force modelling (EGM96 gravity, NRLMSISE-00 drag, solar radiation pressure, third-body perturbations, GCRF/ITRF frame transformations) across two objects of contrasting eccentricity
- Achieved orbit fit improvements of 18–163× during training and up to 617× during validation over conventional Two-Line Element (TLE) propagation, demonstrating operationally significant gains in debris position knowledge
- Compared fixed vs. estimated atmospheric drag coefficient (C_D) configurations, finding eccentricity governs estimation viability (beneficial for near-circular orbits, degrading medium-term prediction for eccentric debris); quantified the resulting drag/along-track observability trade-off and proposed eccentricity-dependent and multi-station tracking strategies for the ARMA active debris removal concept (wider dissertation group project)

SKILLS

Technical & Design Tools: MATLAB, Python, Creo (PTC), SOLIDWORKS, StarCCM, OpenRocket, GMAT, Microsoft Office Suite (Excel, Word, PowerPoint), Power BI

Transferable Skills: Project Management, Teamwork, Data Analysis, Simulation, Communication, Adaptability

Languages: English (Native), French (Native), Spanish (Conversational)

EXPERIENCE

Engineering & Technology Summer Intern – Rolls-Royce, Compression Systems *Summer 2026*

- Investigating digital validation of electron beam welds, developing a predictive model to estimate weld penetration depth from measured beam parameters against a required threshold, reducing material waste and machine downtime
- Leading a lessons-learned review of a new machining coolant rollout across multiple manufacturing sites, consolidating findings to inform future site-wide deployments
- Shadowed the test support team across 6 engine testbeds spanning developmental and production testing, coordinating test schedules and responding to issues; gained insight into engine test data handling and instrumentation

Engineering Intern – Satellite Applications Catapult *June – August 2025*

- Conducted technical research into GNSS-Reflectometry and GNSS-based SAR (40+ papers), evaluating receiver design, signal processing (DDMs, polarimetry, interferometry), and antenna configurations; authored a 60+ page report analysing mission data (CYGNSS, PRETTY, HydroGNSS) in Python, providing recommendations for next-generation GNSS-R missions
- Presented findings internally, recommending priority commercial applications and assessing technical and market feasibility; designed receiver architecture diagrams and processing flowcharts used in both technical and non-technical presentations
- Evaluated application precision trade-offs across domains (soil moisture, ocean altimetry, cryosphere), identifying 3 novel use cases (peatland monitoring, flood risk, precision agriculture)
- Identified 10+ key UK stakeholders (UKSA, ESA, UKRI, Environment Agency, SSTL, Spire), outlining collaboration pathways; built industry connections through engagement with professionals across the Harwell Campus

PROJECTS

QMSAT – S.C.R.A.P. Project Team Member

June 2025 – Present

- Contributing to the mission concept definition of a 3U CubeSat for tracking and classifying space debris using machine learning, balancing technical feasibility against mass, power, and budget constraints
- Selected to represent QMSAT at ESA's CubeSat Concurrent Engineering Workshop: designed a full CubeSat mission concept in 4 days for a rideshare deployment with EnVision, ESA's upcoming Venus orbiter (certificate available upon request)
- As part of the trajectory & mission analysis team, produced rapid preliminary delta-v budgets in MATLAB under tight iteration cycles, feeding directly into propulsion, thermal, and payload sub-team design decisions
- Gained hands-on exposure to concurrent engineering methods using ESA's COMET software and the trade-offs of collaborative mission design

Competition Team Lead Engineer – UKSEDS National Rocketry Championship *Sep 2025 – Jun 2026*

- Lead a team of 11 other students to design and build a mid-power rocket to a target altitude with a unique payload, following strict guidelines and standards; designed a PCB for flight monitoring and data output
- Applied systems engineering methods to integrate propulsion, avionics, structures, and recovery subsystems; overseeing full-cycle rocket development from component selection and modelling to hands-on assembly & testing

ADDITIONAL EXPERIENCE & ACTIVITIES

Engaging independently with AI safety

August 2026

Invisible College (Works in Progress/Stripe) week-long seminar covering AI among other topics; independent reading on AI governance and catastrophic risk

Support Worker – QMUL Disability and Dyslexia Service

September 2024 – Present

Assisted disabled students with access to university facilities and laboratory activities, facilitating communication for those with difficulties

President – Mile End Music Society

May 2024 – August 2026

Lead one of the largest societies on campus (250 members, 6 ensembles); organised an inter-university Battle of the Bands (300+ tickets sold, Event of the Year, QMSU Awards 2025); Best Performing Arts Society, QMSU Awards 2026, teach piano, guitar, and bass to fellow students

Recipient of the Coachmakers' University Scholarship Award

June 2026

Competitive, interview-based aerospace engineering scholarship awarded by the Worshipful Company of Coachmakers to support undergraduate and postgraduate study